

Lecture 5: Three Puzzles in Asset Pricing

Raul Riva

FGV EPGE

January, 2026

Intro

- The equity premium puzzle
- The risk-free rate puzzle
- The volatility puzzle

The Equity Premium Puzzle

The Equity Premium Puzzle

- Mehra and Prescott (1985) is our reference
- In a traditional general equilibrium environment, how large can the equity premium be?
- This is very different from the statistical shenanigans of APT!
- Here: the SDF will heavily depend on consumption growth
- Back to “ m high $\iff$ low consumption growth”

Some (Annual) Data for the US

Time periods	% growth rate of per capita real consumption		% real return on a relatively riskless security		% risk premium		% real return on S&P 500	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
1889–1978	1.83 (Std. error = 0.38)	3.57	0.80 (Std. error = 0.60)	5.67	6.18 (Std. error = 1.76)	16.67	6.98 (Std. error = 1.74)	16.54
1889–1898	2.30	4.90	5.80	3.23	1.78	11.57	7.58	10.02
1899–1908	2.55	5.31	2.62	2.59	5.08	16.86	7.71	17.21
1909–1918	0.44	3.07	−1.63	9.02	1.49	9.18	−0.14	12.81
1919–1928	3.00	3.97	4.30	6.61	14.64	15.94	18.94	16.18
1929–1938	−0.25	5.28	2.39	6.50	0.18	31.63	2.56	27.90
1939–1948	2.19	2.52	−5.82	4.05	8.89	14.23	3.07	14.67
1949–1958	1.48	1.00	−0.81	1.89	18.30	13.20	17.49	13.08
1959–1968	2.37	1.00	1.07	0.64	4.50	10.17	5.58	10.59
1969–1978	2.41	1.40	−0.72	2.06	0.75	11.64	0.03	13.11

SP500 Returns vs Consumption Growth (USA)

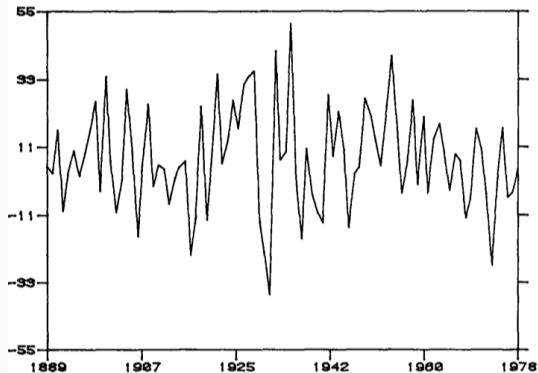


Fig. 1. Real annual return on S&P 500, 1889–1978 (percent).

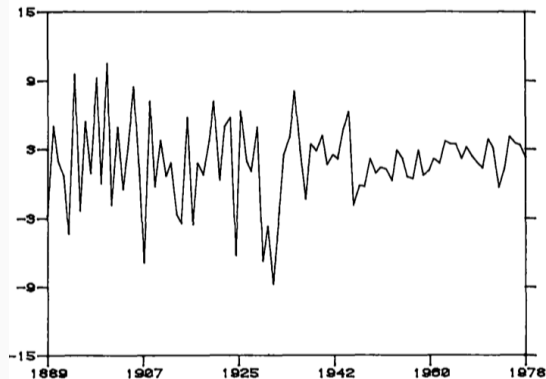


Fig. 2. Growth rate of real per capita consumption, 1889–1978 (percent).

Similar Evidence Across Developed Markets (Quarterly Data)

Country	Sample period	$\bar{r}_e$	$\sigma(r_e)$	$\rho(r_e)$	$\bar{r}_f$	$\sigma(r_f)$	$\rho(r_f)$	$\overline{\Delta c}$	$\sigma(\Delta c)$	$\rho(\Delta c)$
USA	1947Q2–2011Q2	6.85	15.98	0.10	0.72	1.78	0.36	1.74	1.64	0.04
Australia	1970Q1–2011Q2	3.84	20.75	0.03	2.00	2.18	0.55	1.82	1.77	−0.09
Canada	1970Q2–2011Q2	5.47	17.85	0.15	2.07	1.64	0.66	1.69	1.93	0.07
France	1973Q2–2011Q2	7.06	23.10	0.08	2.08	1.53	0.74	1.38	1.80	−0.13
Germany	1978Q4–2011Q2	7.54	23.85	0.04	2.38	1.09	0.40	1.74	4.19	−0.07
Italy	1971Q2–2011Q2	1.51	25.74	0.07	1.86	2.06	0.77	2.18	2.23	0.47
Japan	1970Q2–2011Q2	2.70	21.41	0.09	1.03	1.91	0.29	1.72	2.92	−0.10
Netherlands	1977Q2–2011Q2	8.57	19.76	0.09	2.29	1.42	0.45	1.05	2.21	−0.11
Sweden	1970Q1–2011Q2	8.93	25.16	0.12	1.68	2.23	0.40	1.23	1.81	−0.15
Switzerland	1982Q2–2011Q2	8.14	20.05	0.01	0.87	1.32	0.03	0.75	1.30	−0.22
UK	1970Q1–2011Q2	6.33	19.85	0.10	1.34	2.60	0.46	2.14	2.68	−0.03

Takeaways

- Average return on stocks is high compared to consumption growth across markets
- Consumption growth is way less volatile than stock returns
- Risk-free rates are low and smooth (for developed markets!)
- Stock returns have low autocorrelation
- Consumption growth even has negative autocorrelation sometimes. Italy seems an exception there

The Mehra-Prescott Economy

- They build upon Lucas (1978) (the Lucas' tree model)
- There is representative agent, with preference over consumption streams given by

$$U(c) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \right]$$

where $\beta \in (0, 1)$ and $\gamma > 0$. If $\gamma = 1$, we have log-utility

The Mehra-Prescott Economy

- They build upon Lucas (1978) (the Lucas' tree model)
- There is representative agent, with preference over consumption streams given by

$$U(c) = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \right]$$

where $\beta \in (0, 1)$ and $\gamma > 0$. If $\gamma = 1$, we have log-utility

$$\begin{aligned} \mathbb{E}_t [m_{t+1} R_{t+1}^f] &= 1, & R_{t+1}^f &= \frac{1}{q_t} \\ \mathbb{E}_t [m_{t+1} R_{t+1}^e] &= 1, & R_{t+1}^e &= \frac{p_{t+1} + y_{t+1}}{p_t} \end{aligned}$$

$$\mathbb{E}_t [m_{t+1} R_{t+1}^f] = 1, \quad R_{t+1}^f = \frac{1}{q_t}$$

Household Problem (Sequential)

- Choose $\{c_t, b_{t+1}, s_{t+1}\}_{t \geq 0} \implies$ consumption, bond holdings, and shares of the tree
- State variables: $(b_t, s_t, y_t) \implies$ current holdings of bonds, shares, and dividend
- Prices: q_t (bond), p_t (tree)

$$\begin{aligned} \max_{\{c_t, b_{t+1}, s_{t+1}\}_{t \geq 0}} \quad & \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma} \right] \\ c_t + q_t b_{t+1} + p_t s_{t+1} \leq & b_t + (p_t + y_t) s_t \end{aligned}$$

Notice that marginal utility is given by $u'(c_t) = c_t^{-\gamma}$.

The First Order Conditions

If μ_t is the Lagrange multiplier on the budget constraint at time t , the FOCs are:

$$c_t^{-\gamma} = \mu_t$$

$$q_t \mu_t = \beta \mathbb{E}_t [\mu_{t+1}]$$

$$p_t \mu_t = \beta \mathbb{E}_t [\mu_{t+1} (p_{t+1} + y_{t+1})]$$

The First Order Conditions

If μ_t is the Lagrange multiplier on the budget constraint at time t , the FOCs are:

$$\begin{aligned}c_t^{-\gamma} &= \mu_t \\q_t \mu_t &= \beta \mathbb{E}_t [\mu_{t+1}] \\p_t \mu_t &= \beta \mathbb{E}_t [\mu_{t+1} (p_{t+1} + y_{t+1})]\end{aligned}$$

Define the SDF:

$$\begin{aligned}m_{t+1} &= \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} = \beta \frac{\mu_{t+1}}{\mu_t} \\ \mathbb{E}_t [m_{t+1} R_{t+1}^f] &= 1, \quad R_{t+1}^f = \frac{1}{q_t} \\ \mathbb{E}_t [m_{t+1} R_{t+1}^e] &= 1, \quad R_{t+1}^e = \frac{p_{t+1} + y_{t+1}}{p_t}\end{aligned}$$

Pricing the Tree Recursively

- These equations are not any surprise if you got this far in the class
- If we iterate the equity pricing equation forward and impose the no-bubble condition, we get

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{c_{t+j}}{c_t} \right)^{-\gamma} y_{t+j} \right]$$

Pricing the Tree Recursively

- These equations are not any surprise if you got this far in the class
- If we iterate the equity pricing equation forward and impose the no-bubble condition, we get

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{c_{t+j}}{c_t} \right)^{-\gamma} y_{t+j} \right]$$

- Since the dividend is perishable, $c_t = y_t$ for all t . Then:

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{y_{t+j}}{y_t} \right)^{-\gamma} y_{t+j} \right] = y_t \cdot \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{y_{t+j}}{y_t} \right)^{1-\gamma} \right]$$

Pricing the Tree Recursively

- These equations are not any surprise if you got this far in the class
- If we iterate the equity pricing equation forward and impose the no-bubble condition, we get

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{c_{t+j}}{c_t} \right)^{-\gamma} y_{t+j} \right]$$

- Since the dividend is perishable, $c_t = y_t$ for all t . Then:

$$p_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{y_{t+j}}{y_t} \right)^{-\gamma} y_{t+j} \right] = y_t \cdot \mathbb{E}_t \left[\sum_{j=1}^{\infty} \beta^j \left(\frac{y_{t+j}}{y_t} \right)^{1-\gamma} \right]$$

- Since $q_t = \mathbb{E}_t[m_{t+1}]$, then

$$q_t = \beta \mathbb{E}_t \left[\left(\frac{y_{t+1}}{y_t} \right)^{-\gamma} \right] = \beta \mathbb{E}_t [x_{t+1}^{-\gamma}]$$

Lucas, Robert E. 1978. "Asset Prices in an Exchange Economy." *Econometrica* 46 (6): 1429.

<https://doi.org/10.2307/1913837>.

Mehra, Rajnish, and Edward C. Prescott. 1985. "The Equity Premium: A Puzzle." *Journal of Monetary*

Economics 15 (2): 145–61. [https://doi.org/10.1016/0304-3932\(85\)90061-3](https://doi.org/10.1016/0304-3932(85)90061-3).